Vadim Atlassov

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Research Interests: Physics-Informed Generative Models · Controllable Diffusion · Vision-Language Models · Domain Generalization of Foundation Models

PUBLICATIONS

V. Atlassov, I. Schlattner, Y. Omarov, H. Ju, M.-H. Lee. "Controllable Diffusion-Based Image Generation for Failure Diagnosis of Reinforced Concrete Beam–Column Joints." *Journal of Building Engineering, 128 (2026) 116466. Elsevier.* ***Accepted****. IF 7.4, Q1.*

V. Atlassov, M.-H. Lee. "RECON: A Vision-Language Model for Automated Damage Assessment and Reporting of RC Beam–Column Joints." *In Preparation*. (2026)

I. Schlattner, **V. Atlassov**, A. Abdrakhmanov, M.-H. Lee. "Latent-Aligned Diffusion for Controllable Chest X-ray Synthesis." *Computerized Medical Imaging and Graphics (Under Review)*. (2025)

M.-H. Lee, A. Abdrakhmanov, **V. Atlassov**, I. Schlattner, D. Kim, H. Ju. "A Benchmark Image Dataset of Reinforced Concrete Beam–Column Joint Failures." *Scientific Data, Nature Portfolio (Under Review)*. (2026)

EDUCATION

M.Sc. in Electrical & Computer Engineering, Nazarbayev University 2023
GPA: 3.21/4.00 (Top 10% of Cohort, Graduate Academic Merit Award)

B.Eng. in Electrical and Computer Engineering, EKTU 2020
GPA: 3.71/4.00 (Magna Cum Laude, Rector's Award)

Exchange Program, Brno University of Technology, Czech Republic 2019

RESEARCH EXPERIENCE

Research Assistant April 2025 – Present
Human-Computer Interaction Lab, Nazarbayev University
Supervisor: Prof. Min-Ho Lee

- **Physics-Conditioned Generative Modeling (JBE 2026, Accepted):** Two-stage geometry-conditioned diffusion framework for RC beam–column joint failure synthesis: severity-controlled edge maps (U-Net) condition a frozen Stable Diffusion backbone via multi-scale spatial adapter. KID **0.0628**, SSIM **0.790**; augmentation improved failure classification **80.6%** → **85.8%**, rebar exposure **83.3%** → **95.2%**. Expert realism score **4.12/5**.
- **Controllable Generative Modeling:** Two-stage Latent-Aligned Diffusion for CXR synthesis with structured Text/Coords/Mask conditioning and counterfactual generation (pneumonia → healthy and vice versa). Achieved FID **45.38**, MS-SSIM vs. real **0.736**, and Dice **0.6564**, outperforming state-of-the-art models.
- **Vision-Language Model for Structural Damage Reporting (RECON):** Designed a Mixture-of-Experts adapter and VLM pipeline (DINOv2 → Q-Former → LLaMA 3B) for post-earthquake beam-column joint automated report generation. BLEU-4: **0.706**, CIDEr: **4.243**, ROUGE-L: **0.824**.
- **Clinical Validation & Benchmarking:** Evaluated model reliability via FID (**56.78**) and a blinded radiologist study (Pathology Score: **3.95/5**)

Research Assistant July 2022 – May 2023
ALARIS Robotics Lab, Nazarbayev University
Supervisor: Prof. Almas Shintemirov

- CNN-based 6-DoF pose estimation for UR-5 manipulation in degraded visual environments; **10.2mm MAE** in 3D trajectory alignment.

TECHNICAL SKILLS

Generative: Stable Diffusion, ControlNet, T2I-Adapter, LoRA/PEFT, VAE, Latent Diffusion
Foundation Models: DINOv2/v3, CLIP, BLIP-2, Q-Former, LLaMA, MoE
Infrastructure: PyTorch (DDP/AMP), CUDA, ONNX, TensorRT, Docker, Linux, Git, W&B

HONORS & AWARDS

Presidential Scholarship — Consistent academic excellence, 4 undergraduate years.
Rector's Award — 2nd highest ranking in undergraduate graduating class (EKTU, 2020).
Graduate Academic Merit Award — Top 10% MSc cohort (NU, 2023).

PROFESSIONAL EXPERIENCE

Software Engineer, VEON Ltd (Beeline Kazakhstan) July 2023 – April 2025
Designed and deployed billing systems (11.6M+ users, 99.9% uptime).